

Non–Mortgage Living Expense Estimation Engine

An overview of weeks 4 – 6, ranging from Data Sourcing, Preprocessing, and Model Training

PROJECT CONTRIBUTORS

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K-Means • NIST AI RMF • 279
ZIPs

Executive Summary

🎯 Core Objective

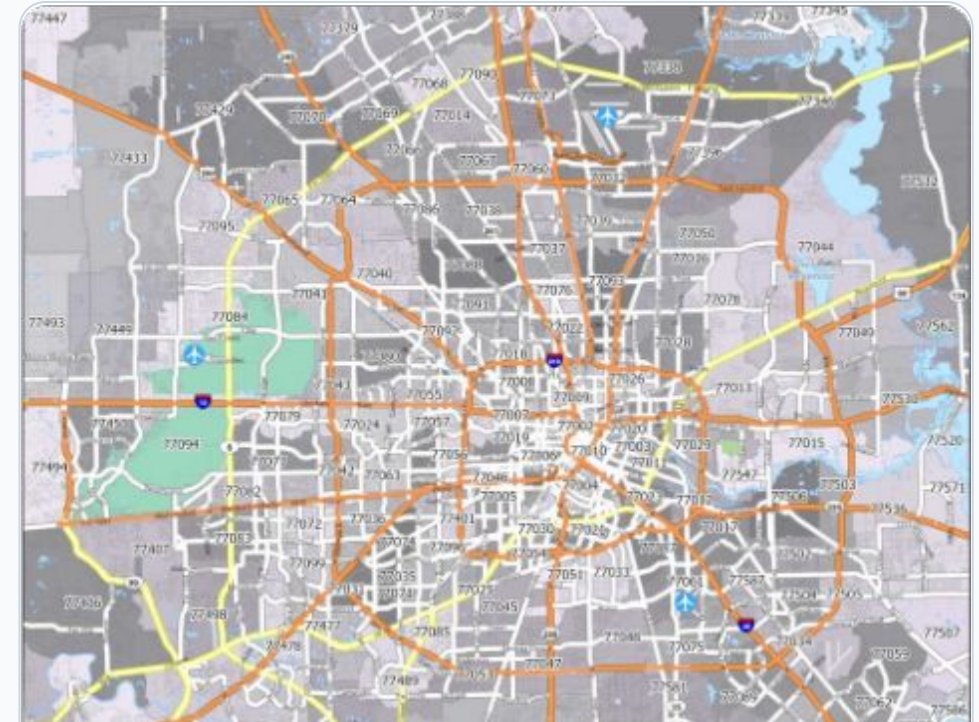
Map and predict hidden non-mortgage living expenses across 279 canonical ZIP codes in Harris, Fort Bend, and Galveston counties.

⚠️ The Problem

Broad regional metrics fail to capture hyper-local utility rate schedules and statutory insurance zones.

✅ The Solution

A localized, data-driven engine that identifies structural cost cliffs and environmental risk exposures.



Houston Metro GIS Spatial Domain

279 Primary ZIPs

The Hybrid Architecture



Pillar 1 • Programmatic

Deterministic Cost Engine

Exact programmatic calculations for published statutory fees, utility rate schedules, and USDA nutritional baselines.

- ✓ Statutory MUD & municipal water rate cards
- ✓ Regulated TDSP electric delivery tariffs
- ✓ County-level USDA food thrift benchmarks



Pillar 2 • Machine Learning

Unsupervised Spatial Clustering

Multidimensional Machine Learning (K-Means) to evaluate geospatial data and identify regional cost-and-risk archetypes.

- ✓ Geospatial FEMA claim density matrices
- ✓ Surge flood and TWIA tier clustering
- ✓ Silhouette-optimized cluster assignment

What this means: We combined fixed statutory billing formulas with machine learning so our system can calculate exact utility rates while discovering hidden geographic risk zones.

Week 4: Multi-Domain Acquisition & Governance



Data Integration

Acquired 9 distinct datasets spanning USPS crosswalks, FEMA NFIP claims, FEMA GIS hazards, CenterPoint/TNMP utility tariffs, and USDA/BLS indices.



Privacy Protection

Strictly anonymized PII from FEMA claims to protect individual policyholders.

PII Masked at Ingestion



Fair Housing Compliance

Adhered to NIST AI RMF 1.0; explicitly omitted demographic, racial, and income features to prevent digital redlining and algorithmic bias.

NIST AI RMF 1.0 Aligned

What this means: We pulled data from 9 official state and federal sources while deliberately excluding race, income, and personal data to ensure compliance and avoid algorithmic bias.

Week 4: Technical Troubleshooting

⚠️ OOM Memory Crashes

The 1.07 GB Texas NFIP dataset exhausted system RAM.



ENGINEERED SOLUTION

Implemented chunked iterative parsing (chunksize=100000) and immediate row-level filtering to reduce memory overhead to ~15 KB.

⚠️ GIS Format Mismatches

FEMA archives contained GeoTIFF rasters rather than vector shapefiles.



ENGINEERED SOLUTION

Developed a spatial fallback module using historical claim density.

⚠️ TCEQ Data Gaps

MUD boundaries lacked USPS ZIP codes.



ENGINEERED SOLUTION

Integrated a spatial fallback crosswalk to correctly map suburban MUD ZIP codes to accurate \$28.50/month base fees.

What this means: When massive flood and utility files threatened to crash our system, we wrote custom chunking and mapping scripts to process the data efficiently.

Week 5: Transformation Pipelines

1

Quality Assurance

Standardized primary keys to 5-digit zero-padded strings and achieved zero missing values across the 279 ZIP code index.

0 Missing Values

2

Feature Engineering

Constructed 20 encoded features balancing physical hazards against regulated infrastructure pricing.

20 Encoded Attributes

3

Pipeline Automation

Designed a Scikit-Learn Column Transformer applying Standard Scaler to continuous metrics and One-Hot Encoder to categorical risk tiers.

ColumnTransformer

What this means: We cleaned and normalized all 279 local ZIP codes into 20 standardized numeric and risk features ready for machine learning.

Week 5: Engineering the Feature Store

Drive Pathing Errors FileNotFoundError

Hyphenated directory mismatches and Colab cache lag triggered .

✓ **Solution: Executed explicit Drive searches and forced cache remounts.**

Vectorization Crashes ValueError

Raw strings (e.g., 'CenterPoint') triggered Scikit-Learn crashes.

✓ **Solution: Redesigned the pipeline to explicitly drop non-encoded string columns.**

Zero-Variance Skew

Static regional constants (e.g., \$18.50 gas base fee) skewed Euclidean distances.

✓ **Solution: Filtered static metrics out of the clustering matrix but retained them for the deterministic engine.**

What this means: We stripped out static fees and non-numeric labels that were causing errors and skewing our distance calculations in Scikit-Learn.

Week 6: Unsupervised Clustering (k = 4)

The Singleton Anomaly (k = 5)

Silhouette: 0.6698

Mathematically peaked but practically failed by isolating an administrative Houston PO Box (77225) into a useless n=1 cluster.

The Optimal Solution (k = 4)

Silhouette: 0.6670

Deployed k=4 to produce robust, highly separated regional archetypes across the entire tri-county study area.

CLUSTER 0

193 ZIPs

Urban Core / Municipal Stable

Established municipal utility grid with baseline non-mortgage living costs.

CLUSTER 1

14 ZIPs

Galveston Mainland / Corridor

Elevated transmission tariffs and transitional coastal weather risk factors.

CLUSTER 2

59 ZIPs

Suburban MUD Growth Corridor

Significant water district fee premiums funding decentralized infrastructure.

CLUSTER 3

13 ZIPs

High-Hazard Coastal Surge Zone

Maximum NFIP claim density coupled with statutory TWIA windstorm mandates.

What this means: Machine learning successfully grouped Greater Houston into 4 real-world expense profiles—from predictable urban centers to high-cost coastal zones.

EDA & Structural Cost Cliffs

Extreme Claim Skew

Flood damage is hyper-localized; the median ZIP saw 286 claims, but major bayou corridors peak at 8,374 claims.

Median ZIP	286 Claims
Bayou Corridor Peak	8,374 Claims

The Suburban MUD Penalty

Suburban enclaves face an automatic \$11.50/month base water penalty over municipal residents (\$28.30 vs. \$16.80).

Municipal Water Base	\$16.80 / mo
MUD Base Rate	\$28.30 / mo

The Coastal Cost Cliff

100% mandatory TWIA windstorm coverage and TNMP's higher electric delivery tariffs generate an immediate >\$200/month structural cost cliff relative to inland zones.

Structural Cost Cliff	+>\$200.00 / mo
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What this means: Moving just a few miles into a suburban MUD or coastal area can automatically trigger hundreds of dollars in extra monthly water and windstorm bills.

Baseline Benchmarks vs. The Granular Engine

NAIVE REGIONAL BASELINE

\$1,684.90

Uniform county averages masking micro-variation

ACTUAL GRANULAR RANGE

**\$1,675.99 –
\$1,891.76**

True localized costs across the 279 ZIP codes

MAX UNDERESTIMATION BIAS

+\$206.86/mo

\$2,482.32/year hidden cost penalty in coastal zones

What this means:

Traditional countywide averages underestimate true household living costs by up to \$206 a month (\$2,482 a year), proving that our hyper-local model is essential.